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Review Article

Critical thinking in mathematics education: A PRISMA-based systematic review

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ABSTRACT

Critical thinking (CT) is widely endorsed in mathematics education; however, its investigation and classroom enactment remain uneven across educational levels, contexts, and methodologies. This PRISMA-guided systematic review examined Scopus-indexed, English-language, open-access journal articles published between 2014 and 2025. Seventeen full texts were assessed from 398 identified records, and 13 included articles met the eligibility criteria. Because one included article was review-based, the corpus is described throughout as "included articles" or "included studies/articles" rather than exclusively as "empirical studies." Within this bounded corpus, the included articles suggest a shift from defining, measuring, and diagnosing CT toward examining questioning, collaboration, statistical literacy, contextual learning, digital tools, teaching practices, and other related factors. This temporal pattern reflects the structure of the reviewed corpus rather than a global development trend. The included articles were primarily situated in higher education and teacher education, with limited attention to primary and upper secondary education. Most articles used qualitative or experimental designs, with fewer mixed-methods, research and development, or review-based designs. Given the small and heterogeneous corpus, a descriptive and thematic synthesis was preferred over a meta-analysis. The synthesis identified four recurring areas: contextual teaching strategies, including ethnomathematics and Realistic Mathematics Education; digital technologies, including multimedia, GeoGebra, and AI-related contexts; collaborative learning through questioning and mathematical dialogue; and teachers' challenges related to knowledge, dispositions, resources, and systemic constraints. Limitations include the small corpus, reliance on a single database and open-access publications, heterogeneous conceptualizations of CT, and the absence of a formal risk-of-bias assessment. Future research should diversify educational levels, geographical contexts, methodological designs, CT conceptualizations, longitudinal frameworks, mixed-methods designs, and studies of teacher agency in technology-rich and culturally responsive mathematics classrooms.

1. Introduction

Critical thinking (CT) has been increasingly recognized as a fundamental competency in mathematics education, as it enables students to move beyond procedural fluency and engage in deeper reasoning, problem solving, and decision-making (Facione, 2011; Leikin & Elgrably, 2020). In mathematics classrooms, students are expected to perform procedures accurately, evaluate arguments, analyze assumptions, justify claims, and make reasonable decisions when solving mathematical problems (Brown, 2016; Su et al., 2015). International assessments such as PISA and TIMSS consistently emphasize mathematical reasoning and CT as essential outcomes for 21st-century learners (OECD, 2022).

Despite its importance, fostering students' CT in mathematics education remains challenging (Peter, 2012; Trisnani et al., 2024). Previous studies have reported that mathematics instruction in various contexts remains characterized by teacher-centered practices, procedural demonstration, and rote memorization, which may limit students' opportunities to engage in argumentation, reflection, and higher-order reasoning (Kusaeri & Aditomo, 2019; Noviyanti et al., 2025; Sehrawat, 2024; Trisnani et al., 2024). Meanwhile, developments in pedagogy, curriculum reform, and digital technologies have created new possibilities for supporting CT through problem-based learning, inquiry-oriented instruction, collaborative learning, ethnomathematical

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contexts, and technology-enhanced mathematics learning (Joldasbaevna et al., 2025; Wang & Abdullah, 2024). However, the extent to which these developments have been empirically examined in mathematics education requires systematic synthesis.

1.1. Literature review

1.1.1. Critical thinking in mathematics education

CT in mathematics is typically defined as the ability to analyze, evaluate, and infer logically in mathematical contexts, while also applying reasoning to novel or complex problems (Ennis, 2011; Facione, 2020). It involves both cognitive skills and dispositions, such as open-mindedness, intellectual persistence, and the willingness to question assumptions (Paul & Elder, 2011). In mathematics education, CT is manifested when students question assumptions underlying mathematical concepts, evaluate multiple solution strategies, justify and communicate reasoning, and flexibly apply mathematics in real-world contexts (Su et al., 2015).

Previous mathematics education studies have examined various strategies for supporting CT, including curiosity-based learning, problem-based learning, contextual learning, and the use of dynamic mathematics software such as GeoGebra (e.g., GeoGebra; Aizikovitsh-Udi & Cheng, 2015; Setyawan et al., 2024). These studies suggest that CT may be supported when students are given opportunities to identify mathematical connections, construct arguments, evaluate solution routes, and reflect on their reasoning. However, the literature remains fragmented in terms of educational level, research method, country of origin, and instructional focus. Therefore, a bounded systematic synthesis is needed to clarify what can be concluded from the specific body of literature captured by explicit search and eligibility criteria.

In this review, all claims are restricted to the corpus of Scopus-indexed, English-language, open-access journal articles published from 2014 to 2025 and identified through the stated search strategy. Therefore, the review does not claim to represent the entire global literature on CT in mathematics education.

1.1.2. The need for a systematic literature review (SLR) on critical thinking in mathematics education

While research on CT in mathematics classrooms is expanding, a bounded systematic synthesis is still needed to clarify how Scopus-indexed, English-language, open-access journal articles from 2014 to 2025 have represented CT in mathematics education. Existing review-oriented discussions have often been narrative, conceptual, or broader in disciplinary scope, and therefore provide limited information about search procedures, eligibility criteria, corpus characteristics, educational levels, methodological designs, and cross-study themes (e.g., Garg et al., 2008; Sampaio & Mancini, 2007). This review does not claim to exhaust the global literature; rather, it maps the subset of literature captured by the stated database, language, access, and search restrictions.

Therefore, a systematic literature review (SLR) is necessary to provide a more rigorous synthesis of research on CT in mathematics education. An SLR enables researchers to map research trends over a defined period, synthesize evidence from studies that meet explicit inclusion criteria, and identify patterns that may not be visible in individual studies or narrative reviews ((Parris and Peachey, 2013); Varsha et al., 2024). In the context of this study, the SLR focuses on empirical research in mathematics education published between 2014 and 2025. This focus is important because CT in mathematics is shaped not only by general theories of reasoning but also by the specific nature of mathematical tasks, representations, argumentation, proof, problem solving, and classroom interaction.

This review has several novel aspects compared to previous review-oriented literature. First, it is specific to the domain of mathematics education, as opposed to discussing the more general idea of CT and education. The second aspect is its synthesis of primary research articles and one review-based article that met the eligibility criteria, rather than relying only on conceptual discussions. For this reason, the term "included articles" is used consistently throughout the manuscript to avoid implying that all articles in the corpus are empirical primary studies. The third aspect is its approach informed by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA), which was applied to ensure that the processes of searching, screening, selecting, and synthesizing were as clear and reproducible as possible. The fourth aspect is that the reviewed studies were analyzed in terms of publication characteristics, educational context, research methods, and thematic content. Considering these novel aspects, it is hoped that the review will shed light on what has been researched, while also identifying what is yet to be explored from an empirical standpoint in CT and mathematics education.

However, the synthesis generated in the present review must be interpreted in the context of the chosen corpus and search strategy. Consequently, it does not claim to cover all existing research on CT in mathematics education. Rather, it offers a systematic description of the characteristics and themes of the empirical studies identified through the review process. This distinction is necessary to prevent the findings from extending beyond the evidence, and broader theories and policy implications are addressed, where appropriate, as contextual observations rather than direct conclusions of the review.

While contributing to research synthesis, this SLR also holds practical relevance for curriculum designers, teacher educators, and policymakers. By identifying recurring instructional approaches, methodological tendencies, and contextual gaps, this review can inform future efforts to more effectively integrate CT into mathematics learning. It can also guide future research by highlighting underrepresented educational levels, geographical contexts, methodological designs, and emerging areas such as technology-supported learning and cross-cultural perspectives (Elizabeth-Thomas & Shende, 2025; Massaro et al., 2016).

Thus, this study aims to systematically review included studies/articles on CT in mathematics education identified through a Scopus-indexed, English-language, open-access journal article search covering 2014–2025. The review addresses four research questions:

1. **RQ1:** How has research on CT in mathematics education evolved from 2014 to 2025?
2. **RQ2:** In what educational contexts has research on CT in mathematics been conducted, and how is it distributed across educational levels and cultural settings?
3. **RQ3:** What research methods are most commonly employed in studies of CT in mathematics education, and what are the emerging methodological trends?
4. **RQ4:** What major themes emerge from research on CT in mathematics education?

2. Methods

2.1. Research design

This study used an SLR approach in alignment with PRISMA. PRISMA was used to support a clear and replicable approach to searching, filtering, selecting, and synthesizing relevant literature (Brooks & McNeely, 2013). This SLR was limited to journal articles published from 2014 to 2025 that addressed CT in relation to mathematics pedagogy, learning, curriculum, assessment, or instructional interventions. The unit of analysis in this review was the included article. The final corpus comprised primary research articles and one review-based article. Consequently, the manuscript uses the terms "included articles" or "included studies/articles" rather than "empirical studies" when referring to the whole corpus.

This review was intended as a descriptive and thematic synthesis

rather than a meta-analysis due to the final corpus being small, methodologically diverse, and comprising studies with various research designs, participants, educational levels, interventions, instruments, and outcome reporting formats. Consequently, the synthesis focused on publication trends, educational contexts, methodological characteristics, and recurrent themes across the included studies.

2.2. Data source and search strategy

The data source for this review was the Scopus database, selected because it provides broad coverage of peer-reviewed international journals across disciplinary areas, including education, social sciences, and mathematics-related research. The systematic search was conducted on September 29, 2025 using the following Boolean search string:

TITLE ("critical thinking") AND TITLE-ABS-KEY ("mathematics") AND PUBYEAR >2013 AND PUBYEAR <2026

The search was intentionally restricted to articles that included *critical thinking* in the title, and *mathematics* in the title, abstract, or keywords. This strategy was applied to increase the likelihood that CT was a central focus of the article rather than a minor or incidental concept. However, this decision also meant that relevant studies using related terms, such as *reasoning*, *higher-order thinking*, *mathematical argumentation*, *problem solving*, or *critical mathematical literacy,* may not have been captured. This limitation should be acknowledged when interpreting the scope of this review.

The search was further limited to peer-reviewed journal articles published in English between 2014 and 2025. An open-access filter was used to ensure access to full-text articles for eligibility assessment and data extraction. Scopus filtering was also applied to retain records relevant to mathematics education. To avoid inconsistencies in reporting, subject-area filtering was treated as part of the search refinement stage rather than as an eligibility decision. The resulting corpus should therefore be understood as a selected body of Scopus-indexed, open-access, English-language journal articles that met the specified search and screening criteria, rather than a complete representation of all global research on CT in mathematics education.

2.3. Study selection and eligibility criteria

Study selection was conducted in two sequential stages. The first involved screening titles, abstracts, and keywords. The second involved full-text eligibility assessment. During the first stage, records were retained when CT was explicitly connected to mathematics education, teaching, learning, or mathematics-related instructional practice. Records were excluded if they addressed CT in a general or cross-disciplinary manner without a substantive focus on mathematics education, mentioned mathematics only peripherally, or provided insufficient information to determine the relevance of CT in mathematics learning or teaching.

The initial search yielded 398 records. As the search was conducted in a single database and export checks showed no identical DOI or title records, no duplicates were removed. Records were then screened based on publication year, document type, language, access status, and relevance to the mathematics education context. After title, abstract, and keyword screening, 17 records were retained for full-text eligibility assessment, which encompassed six criteria (Table 1).

Eligibility was not restricted only to primary empirical studies. Articles were eligible when they provided analyzable information on CT in mathematics education, including primary research articles and review-based articles that explicitly addressed CT in mathematics or STEM-related mathematics contexts. This decision explains the inclusion of Guerra (2024), which was classified as a review-based article rather than as an empirical study with direct participants.

Of the 17 full-text articles assessed, 13 met all inclusion criteria and were included in the final synthesis. The final corpus comprised 12 primary research articles and one review-based article. Four articles were excluded at the full-text stage because they did not provide sufficient focus on CT in mathematics education, lacked adequate methodological information, or did not report analyzable information related to CT constructs, instruments, strategies, or outcomes. Fig. 1 summarizes the complete process of identification, screening, eligibility assessment, and inclusion in a PRISMA flow diagram.

Table 1
Specific criteria for inclusion (study-level checks).

No.	Criterion
1	The title, abstract, or keywords explicitly refer to CT in a mathematics education context.
2	The document is a peer-reviewed journal article.
3	The full text is accessible (open-access filter applied at the search stage).
4	The study explicitly focuses on CT in mathematics education, rather than treating mathematics as a peripheral context.
5	The full text provides sufficient methodological detail to allow assessment and interpretation of the article's design, data source, synthesis basis, or findings.
6	The study reports analyzable information on CT, such as constructs, measurement instruments, pedagogical approaches, or learning outcomes.

2.4. Screening procedure and reviewer roles

To foster transparency and limit selection bias, the first step in the screening process was to apply a structured eligibility checklist in accordance with the criteria listed in Table 1. The first reviewer examined all records by title, abstract, and keywords. The second reviewer independently examined a sample of records and reviewed all borderline cases. Any disagreements or uncertainties regarding the inclusion criteria were discussed. During the full-text assessment, each article was evaluated against the six eligibility criteria, and reasons for exclusion were noted. This process aimed to retain only those studies closely related to CT in the context of mathematics education and which, in the reviewer's judgment, contained sufficient information for synthesis.

2.5. Data extraction and synthesis

The 13 studies were analyzed using a structured data extraction method. Extracted data included author-year citations, names of participating countries, research context, educational level, research design, number of participants, demographic data, research focus, instruments or data sources, data collection methods, instructional interventions, CT constructs and/or indicators, findings, and implications for the research questions.

The synthesis involved a two-part process. The first was a descriptive synthesis of publication year, educational context, research design, and research methodology. The second was a thematic synthesis. In this phase, the analysis examined recurring issues and themes within the included studies. Once identified, these themes were organized through an iterative process of coding, comparing, and classification across studies. This thematic synthesis addressed the research question concerning the major themes and findings of empirical research on CT in mathematics education. External literature was cited only to provide context for the findings in the discussion and was not included in the synthesis of themes and findings.

Given variations in study designs, instruments, interventions, and outcome constructs, pooled effect sizes were not calculated. For quantitative studies, published statistical findings were reported descriptively rather than synthesized through meta-analytic techniques. This approach ensured that the synthesis remained consistent with the available documented evidence.

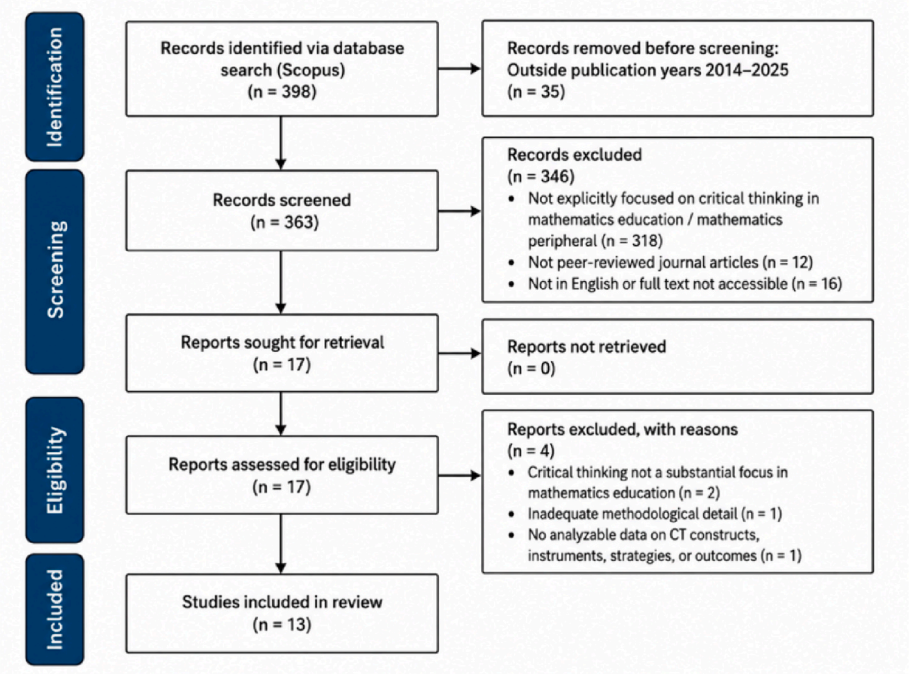


Fig. 1. PRISMA flow diagram for article selection.

2.6. Quality appraisal and risk-of-bias considerations

No formal quality appraisal or risk-of-bias assessment, such as RoB 2, ROBINS-I, or MMAT, was conducted in this review. This decision was made because the corpus was small, methodologically heterogeneous, and included both primary research articles and one review-based article. Instead, methodological sufficiency was addressed at the eligibility stage by requiring each article to provide enough information for interpretation and synthesis. The data extraction process also recorded research design, educational context, participants or data sources, CT constructs or indicators, intervention or data source, and key findings. As a result, the findings of this review should be interpreted as descriptive and thematic patterns within the selected corpus, not as formal judgments of evidence strength or risk of bias. The absence of a formal risk-of-bias assessment is therefore acknowledged as a methodological limitation of this review.

To improve transparency and auditability, a concise evidence table was prepared as Supplementary Table S1. The table summarizes each included article in terms of country, educational level or context, research design, participants or data sources, CT construct or indicator, intervention or data source, and key finding.

3. Findings

3.1. RQ1 – evolution of research publications (2014–2025)

Temporal analysis of the 13 included articles indicates a fluctuating but increasingly diversified pattern of research on CT in mathematics

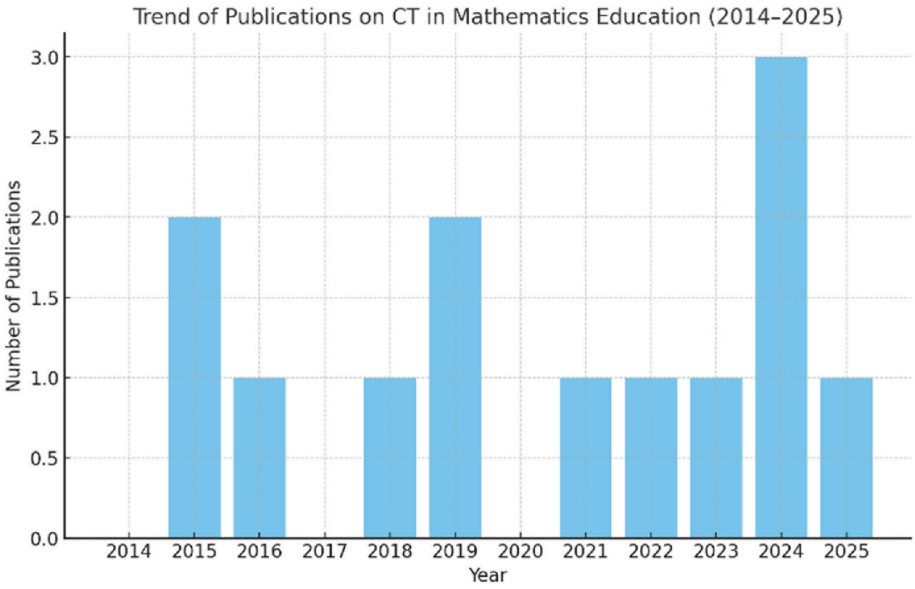


Fig. 2. Publication trend of included articles on CT in mathematics education, 2014–2025.

education from 2014 to 2025. This publication pattern reflects the distribution of the selected corpus identified through this review's search and eligibility criteria rather than a definitive global trend. As shown in Fig. 2, the number of included studies varied by year, with some years represented by one or two publications and a noticeable concentration in the most recent period.

To organize this temporal pattern, studies were grouped into three periods aligned with the broader educational context of the COVID-19 pandemic: pre-pandemic period (2014–2019), pandemic and transition period (2020–2021), and post-pandemic period (2022–2025), as shown in Table 2. These periods are not intended to represent distinct global phases of field development but serve as a contextual and descriptive structure to examine how the focus of the included studies varied across time within this review corpus.

During the pre-pandemic period (2014–2019), the included studies primarily focused on identifying, measuring, and describing CT in formal mathematics education. Some studies used standardized instruments, rubrics, or diagnostic frameworks to assess students' or pre-service teachers' CT levels, while others explored instructional approaches such as Contextual Teaching and Learning (CTL) to support mathematical problem solving and CT (Kurniati et al., 2015; Rasiman, 2015). Sahin et al. (2016) contributed by examining teachers' dispositions toward CT. Chikiwa and Schäfer (2018) explored how questioning in multilingual mathematics classrooms can support justification and reasoning, while Jones (2019) connected CT to statistical literacy by situating mathematical interpretation in authentic social contexts. Overall, studies during this period largely focused on CT measurement, classroom discourse, and early exploration of instructional conditions that support CT.

The pandemic and transition period (2020–2021) included relatively fewer studies in the reviewed corpus, with limited evidence to support broad claims about pandemic-driven shifts in CT research. Nevertheless, they showed continued attention to CT as a socially mediated and reflective process. Calkins et al. (2020) examined Peer-Assisted Reflection (PAR) as a collaborative structure promoting reflection and mathematical communication, while Rott (2021) focused on argumentation and epistemological orientation in mathematical thinking. The period can be interpreted as a transitional phase in which CT was examined through collaborative reflection, argumentation, and reasoning-oriented pedagogies rather than relying on measurement or diagnostic approaches.

The post-pandemic period (2022–2025) contains the largest number of studies in this review corpus and reflects a wider range of themes. These include ethnomathematics, culturally relevant learning, digital learning, multimedia, STEM education, generative AI-related issues, and teacher-related challenges. Utami and Pramudiani (2024) examined culturally grounded learning through a traditional game, while Lestari et al. (2023) and Pratiwi et al. (2022) explored the use of instructional materials, GeoGebra-based resources, and video animations to support students' CT. Guerra (2024) discussed CT in relation to STEM education and generative AI, while Putri et al. (2025) and Trisnani et al. (2024) highlighted teacher-related challenges and dispositions. Overall, this period suggests that post-pandemic CT research paid more attention to the contextual, technological, and teacher-capacity dimensions of CT in mathematics education.

In summary, the reviewed studies suggest a shift within the selected corpus from diagnostic and measurement-oriented work in the pre-pandemic period, to collaborative and argumentation-oriented approaches during the pandemic and transition period, and finally to more contextual, digital, and teacher-capacity-oriented studies in the post-pandemic period. However, this pattern should be interpreted with caution given the limited corpus of 13 studies and its dependence on the search strategy, eligibility criteria, language restrictions, open-access filter, and database selection. Accordingly, the periodization was used as an analytical and contextual framework rather than evidence of a definitive global evolution of CT research in mathematics education. Because the corpus consists of only 13 included articles and is restricted to Scopus-indexed, English-language, open-access journal articles, the temporal grouping should be read as a descriptive organization of the reviewed corpus rather than as evidence of a field-wide historical trajectory.

3.2. RQ2 – educational contexts and distribution

Understanding the educational contexts in which research on CT in mathematics education has been conducted is essential for mapping how CT development is prioritized across learning levels and environments. The distribution of studies across educational levels reflects varying pedagogical goals, challenges, and strategies required to nurture CT skills. This subsection examines the scope of the 13 included studies, identifying the settings in which they were conducted and their implications for curriculum design and teacher professional development.

Table 3 summarizes the distribution of the studies by participants' educational levels. Most studies (53.85%) involved university-level participants, including pre- and in-service teachers. Four studies (30.77%) focused on secondary school students, while one study (7.69%) involved participants from mixed educational levels, such as combinations of junior secondary, senior secondary, and university students. Only one study (7.69%) examined elementary school participants. These findings show that research on CT spans diverse

Table 2
Temporal phases and characteristics of research on CT in mathematics education (2014–2025).

Period	Years	Contextual Framing	Key Focus Areas	Representative Studies
Pre-pandemic period	2014–2019	Research conducted before COVID-19, primarily in conventional classroom and teacher education contexts	CT measurement; CTL and related instructional models; diagnostic studies of CT gaps; questioning strategies; statistical literacy	Kurniati et al. (2015) Rasiman (2015) Sahin et al. (2016) Chikiwa and Schäfer (2018) Jones (2019)
Pandemic and transition period	2020–2021	Research conducted during or around the period of pandemic-related educational disruption and adjustment	Collaborative learning; PAR; mathematical argumentation; epistemological orientation	Calkins et al. (2020) Rott (2021)
Post-pandemic period	2022–2025	Research after the major disruption of the pandemic, with increased attention to contextual, digital, and teacher-capacity dimensions	Ethnomathematics and culturally relevant contexts; digital tools and multimedia; GeoGebra-based resources; AI and STEM education; teacher challenges and dispositions	Guerra (2024) Lestari et al. (2023) Pratiwi et al. (2022) Putri et al. (2025) Trisnani et al. (2024) Utami and Pramudiani (2024)

Table 3
Distribution of studies based on participants' educational levels.

Educational Level	Number of Articles	Representative Studies
University/Pre-service Teachers/ Teachers	7	Putri et al. (2025) Rasiman (2015) Kurniati et al. (2015) Trisnani et al. (2024) Calkins et al. (2020) Sahin et al. (2016) Rott (2021)
Review-based/multiple educational contexts	1	Guerra (2024)
Secondary School	4	Lestari et al. (2023) Pratiwi et al. (2022) Jones (2019) Chikiwa and Schäfer (2018)
Elementary School	1	Utami and Pramudiani (2024)

Note. Guerra (2024) was classified as review-based/multiple educational contexts because it did not involve direct participants at a single educational level.
Total: 13 articles.

educational contexts, with a strong concentration at the university level, thus highlighting the importance of preparing future teachers to foster CT skills across learning environments.

Research on CT in mathematics has been conducted across four main educational contexts, with a strong concentration in higher education. Of the 13 studies, 53.85% focused on university students and pre- or in-service teachers. These studies predominantly explored collaborative and reflective pedagogies, such as PAR, targeting skills such as evaluating mathematical arguments and improving mathematical communication. The findings indicate that such interventions helped students appreciate the roles of peers and reflective practices in developing CT. Moreover, studies involving pre-service teachers highlight the importance of epistemic beliefs and CT dispositions as foundations for designing classroom practice, with a potential multiplier effect: strengthening CT in teacher education is expected to cascade into school-level classrooms.

At the secondary high school level (30.77%), research mainly focused on the development of instructional materials and technology-based interventions (e.g., realistic mathematics education [RME] worksheets, animations, and contextual learning). These approaches aim to foster reasoning, hypothesis testing, and justification skills. One study examined multilingual classrooms and showed that higher-order questioning simultaneously challenges precision and deepens reasoning. However, evidence at this level remains limited compared with its importance for STEM readiness.

Only one included article was classified as review-based/multiple educational contexts. Guerra (2024) discussed CT in STEM education and generative AI across educational contexts rather than reporting data from participants at a single educational level. Therefore, it was not treated as an empirical mixed-level study.

Elementary school research was also underrepresented, accounting for 7.69% of the studies. These studies demonstrated the promise of culturally grounded, play-based learning and immersive media, such as augmented reality, in supporting early CT development by linking children's daily experiences with formal mathematical concepts. However, the limited evidence indicates a clear need for more longitudinal and culturally responsive research.

Geographically, the studies were concentrated yet diverse (see Fig. 3). Indonesia contributed to 54% of the studies, aligned with Merdeka Belajar educational reforms and a robust ethnomathematics tradition, while international studies (South Africa, Wales, the USA) more often used advanced methods (mixed methods, digital tools). Together, they are complementary: local studies offer context-rich and culturally relevant models, while global studies provide scalable frameworks. Evidence tends to cluster where it can shift the system (teacher education), but thins where CT formation begins (early grades) and where pathways crystallize (upper secondary). A coherent agenda should (i) develop spiral CT across schooling, (ii) upskill teachers in digital and culturally responsive CT, (iii) align assessment and policy with CT goals, and (iv) build global–local partnerships to test scalable, context-sensitive designs.

Overall, this educational distribution demonstrates that the included articles in this corpus predominantly address university and teacher education contexts. While secondary school contexts are represented, primary school and cross-level developmental research remain limited. This should be interpreted as a pattern within the reviewed corpus, not as a definitive conclusion about the global field of CT research in mathematics education.

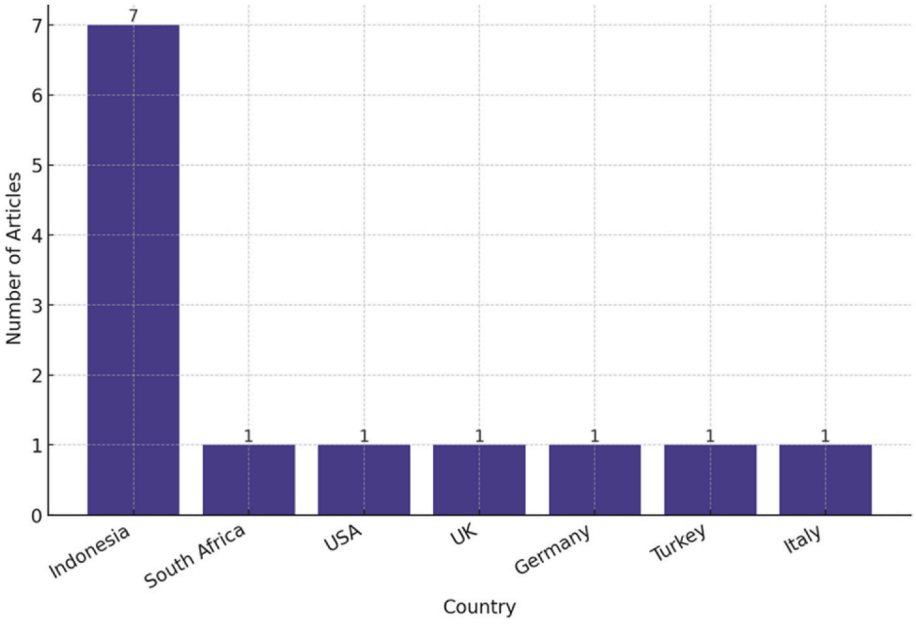


Fig. 3. Distribution of included articles by country.

3.3. RQ3 – research methodology

The distribution of the selected studies sheds light on the investigation of CT in mathematics education. The 13 included articles employed various methodological approaches, including qualitative, descriptive, and exploratory designs; quantitative and experimental designs; mixed methods; research and development; and one review-based SLR article. Table 4 shows that the majority of the studies—five out of 13—were qualitative, descriptive, or exploratory. These studies generally examined CT as a classroom-, cognitive-, or teacher-related phenomenon. They used methods such as interviews, observations, document analysis, analysis of students' work, and qualitative interpretations of classroom practices. Their contribution lies in providing detailed accounts of how CT is understood, enacted, or constrained in specific mathematics education contexts. However, this context-specific nature limits the extent to which findings can be generalized beyond each study's setting.

Various methods can be employed to navigate the complexities of studying CT in mathematics education. CT involves both externally observable actions, such as problem-solving performance, and cognitive functions including thinking, reasoning, inference, and reflection. Capturing this duality requires a range of methods including, but not limited to, in-depth qualitative research, controlled studies, and integrative reviews.

The largest share of studies (38%) were qualitative, descriptive, or exploratory and examined CT as a lived classroom experience. They generally involved small samples and relied on interviews, classroom observations, document analysis, and think-aloud protocols. They highlighted how students and teachers engage in CT through proof strategies, discourse, and contextual learning. For example, one ethnographic study showed how the traditional game *Damdas 3 Batu* helped Indonesian students connect cultural practices with mathematical reasoning. While these studies provide nuanced insights and support theory building, their context-specific nature limits broader generalizability.

Quantitative experimental studies (31%) investigated intervention effects using larger samples and statistical testing (e.g., t-tests, ANOVA). They evaluated how specific teaching methods, such as video animations or contextual teaching, improved students' CT in mathematics. These designs establish causal claims about "what works," but often emphasize measurable outcomes at the expense of capturing deeper processes of reasoning and reflection.

Only one study employed an R&D approach (8%) by creating new resources and instructional materials, including digital worksheets combining ethnomathematics and GeoGebra. The practical contributions of these studies lie in iterative cycles of design, validation, and testing, which may explain their limited number, as these activities are resource-intensive.

Mixed-methods studies (15%) combined qualitative depth with quantitative breadth through an integrated approach. For instance, one study qualitatively analyzed teacher-questioning patterns and subsequently scaled up the analysis using survey data to explore the patterns. These designs are particularly suited to addressing the complexities of CT, reflecting the view that no single approach can fully capture all dimensions of the phenomenon.

Finally, one study (8%) employed an SLR, synthesizing findings across sources to identify trends, gaps, and future research directions. Although scarce, such reviews are important for consolidating evidence and informing policy and practice.

Overall, the methodological distribution suggests that research on CT in mathematics education within the selected corpus was methodologically diverse but uneven. Qualitative and experimental studies dominated, whereas mixed-methods, R&D, and review-based articles were less frequent. Given the small size and heterogeneity of the corpus, these findings should be interpreted as descriptive patterns rather than stable methodological trends across the broader field.

Table 4
Distribution of studies across five major methodological categories.

Research Methodology	Number of Articles	Representative Studies
Qualitative/Descriptive/Exploratory	5	Putri et al. (2025) Utami and Pramudiani (2024) Pratiwi et al. (2022) Rasiman (2015) Trisnani et al. (2024)
Quantitative Experimental	4	Kurniati et al. (2015) Jones (2019) Sahin et al. (2016) Rott (2021)
Research and Development (R&D)	1	Lestari et al. (2023)
Mixed Methods	2	Calkins et al. (2020) Chikiwa and Schäfer (2018)
Review-based article/Systematic Literature Review (SLR)	1	Guerra (2024)

Note. The review-based article was retained because it met the eligibility criteria and provided analyzable information on CT in STEM-related mathematics education; however, it was not treated as a primary empirical study.

3.4. RQ4 – emerging themes

Four recurring themes were identified through the synthesis: integration of contextual approaches, utilization of digital technology, collaborative learning strategies, and teacher challenges and professional dispositions. These themes are summarized in Table 5.

3.4.1. Theme 1: integration of contextual approaches

Several studies in the corpus suggested that contextualized mathematics learning can provide meaningful opportunities for CT development. These studies used cultural contexts, RME, CTL, local games, and authentic social issues to connect mathematical reasoning with students' experiences. Utami and Pramudiani (2024) demonstrated how the Betawi traditional game *Damdas 3 Batu* supported students' interpretation, analysis, and evaluation skills. Lestari et al. (2023) showed that RME-based worksheets encouraged modeling, multiple solution pathways, and reflection. Kurniati et al. (2015) confirmed that CTL principles such as inquiry, questioning, and reflection improved students' mathematical CT skills. Similarly, Jones (2019) illustrated how embedding statistics in authentic social issues through a pilot course

Table 5
Emerging themes from the thematic synthesis.

Main Theme	Description	Representative Studies
Integration of Contextual Approaches	CT is fostered through culturally responsive pedagogies such as ethnomathematics, RME, and the use of local games.	Jones (2019) Kurniati et al. (2015) Lestari et al. (2023) Utami and Pramudiani (2024)
Utilization of Digital Technology	Digital tools such as video animations, AI-driven apps, and online platforms act as mediators for CT development.	Guerra (2024) Pratiwi et al. (2022)
Collaborative Learning Strategies	CT is activated through questioning, PAR, group problem solving, and structured discussions.	Calkins et al. (2020) Chikiwa and Schäfer (2018)
Teacher Challenges and Professional Dispositions	Barriers teachers face in understanding, valuing, and implementing CT-centered pedagogy.	Putri et al. (2025) Rasiman (2015) Rott (2021) Sahin et al. (2016) Trisnani et al. (2024)

shifted learners' attitudes and strengthened CT. Across these studies, contextual approaches appeared to support CT by making mathematical problems more meaningful, interpretable, and open to reasoning. However, the evidence remains limited in size and context. Therefore, this theme should be interpreted as a recurring pattern in the selected studies rather than conclusive evidence that contextual approaches consistently improve CT.

3.4.2. Theme 2: utilization of digital technology

The second theme concerns the role of digital technology in supporting or framing CT in mathematics education. In the reviewed studies, technology appeared in the form of video animation, GeoGebra-based resources, digital worksheets, online learning environments, and generative AI-related discussions. Pratiwi et al. (2022) found that video animations during pandemic-era teaching improved students' ability to evaluate solutions in social arithmetic. Guerra (2024) extended this perspective by examining the role of generative AI in STEM, arguing that CT is essential for the ethical and reflective use of technology. These studies suggest that digital tools can mediate visualization, interaction, feedback, and reflection. However, the evidence in the corpus does not support broad claims that digital technology is a universally effective solution for CT development. Issues such as access, teacher readiness, instructional design, and students' ability to evaluate digital outputs remain important considerations.

3.4.3. Theme 3: collaborative learning strategies

The third theme relates to collaborative learning, dialogue, questioning, and peer reflection. In this corpus, CT was not treated solely as an individual cognitive skill but also as something that can emerge through social interaction and mathematical communication. Chikiwa and Schäfer (2018) showed that teacher questioning in multilingual classrooms, when effectively scaffolded, activated deeper reasoning and critical engagement. Calkins et al. (2020) demonstrated that PAR shifted students' focus from simply producing answers toward evaluating reasoning and communicating arguments more effectively. These studies indicate that collaborative structures may support CT by requiring students to articulate, defend, revise, and evaluate mathematical ideas. However, this theme is represented by a small number of studies; therefore, further empirical work is required before making stronger claims about the comparative effectiveness of collaborative strategies.

3.4.4. Theme 4: teacher challenges and professional dispositions

The fourth recurring theme concerns teacher-related challenges and professional dispositions. Several studies showed that CT development in mathematics education depends not only on students or learning materials, but also on teachers' understanding, beliefs, dispositions, and capacity to design CT-oriented instruction. Sahin et al. (2016) showed that while teachers reported strong professional values, their CT dispositions remained low. Putri et al. (2025) found that pre-service teachers' understanding of CT was narrow and often limited to basic interpretations. Rasiman (2015) identified wide variations in CT levels among pre-service teachers, underscoring difficulties in applying analysis, evaluation, and logical justification. Trisnani et al. (2024) reported systemic barriers such as time constraints, limited resources, and uncertainty about expected outcomes among Indonesian teachers. Rott (2021) further highlighted that students' CT depends less on epistemological stance than on the sophistication of their argumentation, an area in which many pre-service teachers struggle. Collectively, these studies suggest that teacher education and professional development are central to CT implementation in mathematics classrooms. However, the evidence also indicates that teacher-related challenges are complex and context-dependent. Therefore, professional development should not be treated as a simple solution but as one component of a broader instructional and institutional effort to support CT-oriented mathematics learning.

In summary, this review's findings indicate that the selected corpus of 13 studies is characterized by four main patterns. First, publication patterns show increasing thematic diversity over time, particularly in relation to contextual learning, technology, collaboration, and teacher-related challenges. Second, the studies are concentrated at the university and teacher education levels, while elementary and mixed-level contexts remain underrepresented. Third, the corpus is methodologically diverse but dominated by qualitative and experimental studies, with fewer mixed-methods, R&D, and review-based studies. Fourth, the thematic synthesis identified four recurring themes: contextual approaches, digital technology, collaborative learning, and teacher challenges.

4. Discussion

4.1. RQ1: evolution of research publications (2014–2025)

The findings for RQ1 show that research on CT in mathematics education within the selected corpus developed unevenly between 2014 and 2025. Rather than indicating a definitive global increase in publication activity, this pattern should be interpreted as a descriptive trend within the 13 studies included in this review. Although the number of included articles remained small across the period of study, the thematic focus became increasingly varied, particularly in recent years. This pattern suggests that CT in mathematics education has increasingly been examined not only as an individual cognitive skill, but also in relation to classroom discourse, contextual learning, technology integration, and teacher capacity. This interpretation remains bounded by the review's scope. The observed diversification does not demonstrate a universal development of CT research in mathematics education; rather, it shows how the 13 Scopus-indexed, English-language, open-access articles included in this review are distributed across time and theme.

The three periods presented in Section 3 were defined in relation to the broader educational context of the COVID-19 pandemic: pre-pandemic (2014–2019), pandemic and transition (2020–2021), and post-pandemic (2022–2025). This periodization helps clarify how the focus of the included studies has changed; however, it should not be interpreted as evidence of a universal shift in CT research in mathematics education.

The studies included in the corpus under review primarily focused on the identification, measurement, and description of CT in mathematics education during the pre-pandemic period (2014–2019). These studies laid the foundation for subsequent CT research by providing insights into students' and teachers' CT levels, predispositions, instructional challenges, and classroom environments that support CT. This focus aligns with early CT research, which emphasizes analysis, assessment, inference, and reflective judgment as key components of reasoning (Ennis, 2011).

Some studies during 2014–2025 also examined CT in relation to instructional models, questioning approaches, and statistical literacy instruction. As such, the studies in the post-pandemic period did not focus solely on CT measurement, but also began exploring how mathematical reasoning, justification, and contextual interpretation could support CT. However, the findings from this period were limited in scope and therefore do not provide a strong basis for claims of widespread CT-related instructional change in classrooms.

Between 2020 and 2021, during the pandemic and transition period, fewer studies were identified in the analyzed corpus. Consequently, this period does not substantiate extensive claims regarding shifts in the focus of CT research. However, the studies from this period point to a sustained focus on CT as a socially constructed and reflective activity. PAR, collaborative mathematical communication, argumentation, and epistemological orientation were highlighted as important elements. These studies suggested that CT in mathematics is partly supported by students' ability to justify, assess, and revise their reasoning within peer-interactive settings. This perspective aligns with calls for pragmatic approaches in educational research, particularly in complex domains

that balance research outcomes with learning processes (Creswell & Plano Clark, 2018). Nevertheless, due to the limited number of studies during this period, this focus should be interpreted as an emerging rather than an established trend.

From 2022 to 2025, the reviewed studies reflected an increased focus on context, culture, STEM, digital technology, AI, and teacher capacity in relation to CT. Studies on ethnomathematics and RME integrated local games and cultural practices into the curriculum and connected students' lived experiences with formal mathematical reasoning. This aligns with culturally responsive and sustaining pedagogy, as discussed by Bishop (2013) and Paris and Alim (2017).

Digital media featured prominently in these studies, with video animations and GeoGebra examined as tools for visualization and reasoning. This interpretation is also consistent with the TPACK view, which argues that the successful technology integration depends on the interaction among technological, pedagogical, and content knowledge (Mishra & Koehler, 2006). Additionally, early work on AI and STEM positioned CT as essential for evaluating information and outputs, as well as for the reflective use of technology (Luckin & Holmes, 2016). Nevertheless, these reflect broader contextual perspectives rather than direct evidence from the analyzed studies.

Taken together, the selected studies suggest a shift within the reviewed corpus from diagnostic and measurement-oriented research in the pre-pandemic period, to collaborative and argumentation-oriented research during the pandemic and transition periods, and then toward more contextual, digital, and teacher-capacity-oriented studies in the post-pandemic period. However, this shift should not be overstated. The final corpus comprised only 13 studies, many of which were concentrated in specific geographical and institutional contexts. Therefore, the findings are best understood as characteristics of the reviewed sample rather than evidence of a mature or globally representative research trajectory. Future studies should strengthen this evidence base through broader search strategies, larger samples, longitudinal designs, and mixed-methods studies that connect learning outcomes with the mechanisms through which CT develops in mathematics classrooms (Johnson & Onwuegbuzie, 2004).

4.2. RQ2: educational contexts and distribution

The results for RQ2 suggest that the sampled studies were unevenly distributed across educational levels. The most notable representation occurred in university and teacher education contexts, particularly involving pre- and in-service teachers. This distribution indicates that the literature places strong emphasis on teacher education and higher education as key domains for cultivating CT in mathematics education. This focus is justified, as CT dispositions, beliefs, and practices in teacher education and classroom instruction influence the enactment of CT in school settings.

However, this emphasis also reveals an imbalance in the evidence base. Elementary- and mixed-level studies were minimally represented, whereas secondary school studies appeared in a smaller but still notable proportion. This uneven distribution limits the extent to which this review can explain how CT develops across educational stages. Although CT is widely recognized as an important educational outcome (Facione, 2011; OECD, 2022; UNESCO, 2017), the reviewed corpus does not provide sufficient longitudinal or developmental evidence to explain how CT emerges from early schooling through higher education.

Although teacher education was emphasized in the reviewed studies, this should be interpreted with caution. Studies involving both pre- and in-service teachers argued that professional attitudes, epistemological beliefs, and pedagogical understanding are important for teaching mathematics with a focus on CT. For instance, Putri et al. (2025), Rasiman (2015), Sahin et al. (2016), Rott (2021), and Trisnani et al. (2024) highlighted various aspects of teacher readiness and barriers. The TPACK framework is also useful for analyzing teacher readiness for CT in technology-rich environments, as it addresses the intersection of technological, pedagogical, and content knowledge (Mishra & Koehler, 2006). Nonetheless, it would be unwise to conclude that the findings from teacher education studies translate directly into teaching practice without further evidence.

At the school level, the studies described support for CT through contextual, technological, and dialogic approaches. Secondary-level studies examined video animation, statistical literacy, and teacher questioning, while elementary-level studies focused on context grounded in games. These studies help illuminate how CT is integrated into mathematics learning. However, because such studies are few, it is difficult to support claims of positive impact across educational levels. The corpus strongly suggests the need for more research at the elementary and senior secondary school levels, where learners' reasoning, justification and problem-solving habits may develop in qualitatively different ways.

The geographical distribution of the reviewed studies was less balanced than school-level distribution. A large number of studies were conducted in Indonesia. This concentration is not necessarily negative, as it reflects culturally and contextually relevant examples of mathematics education. However, it limits the comprehensiveness of the review. Hence, the results cannot be interpreted as global trends in CT research in mathematics education. Rather, they represent trends within the selected corpus, which was influenced by database selection, language, search terms, open-access filters, and eligibility criteria. Further research would benefit from broader international coverage, particularly from underrepresented regions.

4.3. RQ3: research methodology

The methodological distribution of the reviewed studies shows that CT in mathematics education has been examined using diverse research designs. Within the selected corpus, qualitative, descriptive, and exploratory studies were the most common, followed by quantitative experimental studies, mixed methods, R&D, and one review-based study. This distribution indicates methodological diversity, but it should not be interpreted as a stable trend across the broader field due to the small number of included studies.

Qualitative studies were prominent in the corpus because they were well suited to describing CT processes in mathematics classrooms, including how learners construct arguments, justify claims, participate in mathematical discourse, and connect cultural contexts with mathematical reasoning. They describe how learners construct arguments and justifications, and how they collaborate using ethnographic or case study methods. For example, one study used ethnographic methods to demonstrate how the game *Damdas 3 Batu* in a class of 30 elementary students supported CT development in a culturally diverse classroom context. Similarly, Fitriyah et al. (2025) and Pradhan (2025) highlighted the usefulness of qualitative approaches for integrating CT in mathematics, as they included elements of cultural practices. However, because many qualitative studies involved relatively small and context-specific samples, their findings have limited external validity and should not be generalized beyond the settings in which they were conducted.

R&D and mixed-methods models integrate theory and practice through iterative designs to create GeoGebra-based worksheets and culturally integrated modules. Both approaches are conceptually strong and practically useful; however, collaborative partnerships between researchers and educators are critical. Kasti and Jurdak (2017) demonstrated the effectiveness of collaborative GeoGebra-based interventions in mathematics education. Penuel et al. (2011) described the value of design-based implementation research in addressing persistent practice-based issues. Kusi and Bonyah (2025) demonstrated measurable gains in student performance through culturally integrated modules using R&D designs. Latifa et al. (2025) noted the limited number of SLRs in this area and highlighted the need for such reviews to support curriculum policy and development processes. Although these designs show

potential, R&D and mixed-methods approaches remain at an early stage of development, with a lack of systematic research to address persistent practice-based issues and sustain implementation processes.

Overall, the methodological pattern suggests that research on CT in mathematics education requires approaches that capture both outcomes and processes. Qualitative research provides depth, quantitative research provides measurable evidence, and mixed-methods or design-based studies help connect learning mechanisms with instructional effects. Future studies should report methodological procedures more transparently, including instruments, validity evidence, intervention design, and statistical results where relevant. This would improve the comparability of findings and enable future reviews to conduct statistical synthesis where appropriate.

4.4. RQ4 – emerging themes

The thematic synthesis revealed four themes related to the selected corpus: contextual approaches, digital technology, collaborative learning strategies, teacher-related challenges and professional dispositions. These themes indicate that CT in mathematics should not be treated as an isolated cognitive skill of the learner. Rather, it is defined in relation to task, culture, technology, classroom dynamics, and the teacher. Nevertheless, these themes should be viewed as trends based on the 13 studies reviewed and not as exhaustive categories in the field.

The four themes should therefore be interpreted as synthesis categories generated from the included corpus, not as a complete taxonomy of CT research in mathematics education. This distinction is important because the corpus is small, database-specific, English-language only, open-access only, and methodologically heterogeneous.

The first theme concerns contextual approaches, including ethnomathematics, RME, CTL, local games, and authentic problem situations. Utami and Pramudiani (2024) and Lestari et al. (2023) demonstrated how local games and real-world modeling tasks foster CT by connecting students' lived experiences with formal reasoning. Supporting this, Rosa and Orey (2013) found that culturally relevant pedagogy enhanced students' intellectual and social learning through cultural referents, while Nasrum et al. (2025) confirmed a significant rise in ethnomathematical studies, reflecting the growing recognition of culture-based mathematics learning. Abbas et al. (2025) further demonstrated that the success of RME depends on the integration of local cultural values and students' daily life contexts. However, Hadi (2002) and Kusi and Bonyah (2025) noted that large-scale implementation faces challenges such as limited teacher training and curriculum adaptation, consistent with concerns about testing pressures and teacher readiness that hinder sustainable integration, thereby emphasizing the need for curriculum reform and continuous professional development.

The second theme relates to digital technology and its application in the selected studies as a resource for visualization, interaction, feedback, and learning assistance. Pratiwi et al. (2022) focused on video animation, Lestari et al. (2023) included GeoGebra-based instructional resources, and Guerra (2024) analyzed the role of CT in relation to generative AI and STEM education. These studies suggest that technology can foster CT when paired with meaningful mathematical activities and teaching strategies. The TPACK model supports this view, as it argues that technology enhances learning when integrated with pedagogy and content (Mishra & Koehler, 2006). However, technology should not be considered inherently beneficial. Factors such as teacher preparedness, instructional design quality, and learners' ability to assess digital products remain critical.

The third theme concerns collaborative learning strategies. Chikiwa and Schäfer (2018) and Calkins et al. (2020) suggested that questioning, PAR, and structured dialogue provide opportunities for students to explain, justify, challenge, and revise mathematical ideas. This theme indicates that CT in mathematics may develop through social interaction, not only through individual cognition. However, because this theme was represented by a small number of studies, stronger claims about collaborative learning require further empirical support across different educational levels and classroom contexts.

The fourth theme concerns teachers' challenges and professional dispositions. Several studies showed that the implementation of CT-oriented mathematics instruction depends on teachers' understanding of CT, confidence, pedagogical resources, time constraints, assessment expectations, and professional beliefs. Putri et al. (2025), Rasiman (2015), Sahin et al. (2016), Rott (2021), and Trisnani et al. (2024) collectively suggested that teacher readiness is a central issue in CT implementation. This finding is important because even well-designed contextual or digital materials may have limited impact if teachers are not prepared to facilitate reasoning, argumentation, reflection, and evaluation in mathematics classrooms.

Overall, the four themes indicate that CT in mathematics education should be treated as a multidimensional phenomenon, incorporating the design of learning activities, cultural elements, technology, classroom interactions, and teacher capability. The literature reviewed reveals that CT in mathematics education has promising potential, although the evidence is uneven and scarce. Accordingly, the aim of this review is not to make general claims about global applicability but to describe the characteristics, methodologies, and themes of the selected literature. The available literature calls for more extensive, varied, and methodologically rigorous studies, especially in educational levels that remain underexplored or neglected.

5. Conclusion

This systematic review examined included studies/articles on CT in mathematics education published between 2014 and 2025 and identified through a specified Scopus-based search strategy. The final corpus comprised 12 primary research articles and one review-based article. Therefore, the findings describe patterns within the reviewed corpus rather than the full global literature on CT in mathematics education.

The reviewed literature includes more discussions on higher education and teaching education than in elementary and higher secondary education. This suggests that the current body of literature does not sufficiently demonstrate the progressive development of CT across educational levels. From a methodological standpoint, the cited literature exhibits considerable diversity and includes qualitative, quantitative, experimental, and mixed-methods studies, as well as development-focused research and literature reviews. However, the limited amount of literature, variety of study designs, and different ways in which the outcomes were reported precluded statistical synthesis or meta-analysis. Thus, this review relied on a descriptive and thematic synthesis rather than a pooled analysis to demonstrate the effectiveness of interventions.

Four recurring themes emerged from the synthesis: contextual approaches, such as ethnomathematics, RME, and authentic problem contexts; the use of digital technology, including multimedia, GeoGebra-based resources, and AI-related contexts; collaborative learning strategies, such as questioning, PAR, and mathematical dialogue; and teacher-related challenges involving knowledge, beliefs, dispositions, resources, and systemic constraints. These themes suggest that CT in mathematics education is shaped not only by students' individual cognitive abilities but also by instructional design, cultural relevance, technological mediation, classroom interaction, and teacher capacity.

This review aimed to delineate the empirical literature within the domain of mathematics education and CT. It organized data from the reviewed studies, focusing on educational contexts, research methods, covered themes, and observed patterns across studies. The pre-pandemic, pandemic and transitional, and post-pandemic periods used in this review reflect the order in which the studies were reviewed, and do not represent distinct historical periods or global trends in the field. Similarly, the four themes identified in this review were constructed from the included studies and do not define or limit the scope of CT research in mathematics education.

This study has several limitations. First, the corpus was small, comprising 13 studies, and was influenced by the search method, database choice, use of the English language, open-access filters, and inclusion criteria. Second, this review did not conduct a formal quality appraisal or risk-of-bias assessment. Although methodological sufficiency was considered during eligibility screening and data extraction, the review did not formally rate the quality, evidence strength, or risk of bias of each included article. Consequently, the synthesis should be read as a descriptive and thematic mapping of the corpus rather than as a formal evaluation of evidence quality. Third, because most studies originated from Indonesia, the geographical imbalance of the reviewed studies limits the generalizability of the findings. Fourth, the included studies used differing definitions, measurements, and assessments of CT, which complicated cross-study comparisons. Fifth, few longitudinal studies have been conducted, further limiting the strength of claims regarding the sustainability of some interventions. Equity was minimally reported, particularly in relation to gender, language, income level, and access to technology.

Consequently, future research should expand the evidence base by incorporating greater geographical, educational, and methodological diversity. CT research should investigate the development of CT through age-appropriate, culturally responsive, and mathematically meaningful activities. CT assessment should be more rigorous, transparent, and conducive to cross-study comparison. To better understand the role of CT in mathematics classrooms, further research on learning contexts, digital technologies, collaboration, and teachers' professional development is needed. It is also important to examine digital divides and the role of AI in mathematics education while addressing ethical, pedagogical, and epistemological concerns.

Overall, this review suggests that CT in mathematics education is a promising but still unevenly developed area of empirical research. Strengthening the field will require more robust evidence, clearer conceptualization of CT, transparent methodology, and sustained attention to the classroom, cultural, technological, and teacher-related conditions that shape students' opportunities to think critically in mathematics.

Data availability statement

The data supporting this systematic review are available within the article and its Supplementary Material, including list of included articles, and extracted evidence table. Additional extraction records are available from the corresponding author upon reasonable request.

Ethics in publishing statement

I testify on behalf of all co-authors that our article submitted followed ethical principles in publishing.

All authors agree that:

This research presents an accurate account of the work performed, all data presented are accurate and methodologies detailed enough to permit others to replicate the work.

This manuscript represents entirely original works and or if work and/or words of others have been used, that this has been appropriately cited or quoted and permission has been obtained where necessary.

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All authors have been personally and actively involved in substantive work leading to the manuscript and will hold themselves jointly and individually responsible for its content.

Declaration of the use AI

During the preparation of this revised manuscript, the authors used ChatGPT by OpenAI to assist with language editing of text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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CRediT authorship contribution statement

Lathiful Anwar: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Cholis Sa'dijah:** Investigation, Methodology, Visualization, Writing – review & editing. **Ety Tejo Dwi Cahyowati:** Conceptualization, Investigation, Methodology, Writing – review & editing. **Indrani Eka Prastya Zahroh:** Data curation, Investigation, Methodology, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ssaho.2026.103284.

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